The cognitive turn in experimental pragmatics: Towards a computational cognitive science of communication

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Abstract

concrete.

1 Experimental pragmatics and computational modeling

The linguistic subfield of pragmatics studies how language is used for efficient communication and how communicated meaning is modulated, if not created, in a particular context by systematic, context-general patterns of use. Its origins lie in the philosophy of language, in particular the analytical tradition of philosophy, in work on topics like *speech-acts* (Austin, 1962; Searle, 1969), *conversational implicature* (Grice, 1975), or *presupposition* (Stalnaker, 1973; Karttunen, 1973). Much early work on these topics remained invested in a logical-formal approach (Gazdar, 1979; Kadmon, 2001), but the early 2000's brought a deep and empowering incision: a widely adopted and ultimately very successful *experimental turn* (Noveck and Sperber, 2004; Noveck, 2018), in parallel to similar empirical re-orientations in other hitherto very analytically minded subfields of linguistics like formal syntax (Sprouse, 2007, 2023) and formal semantics (Meibauer and Steinbach, 2011; Cummins and Katsos, 2019).

The experimental turn in pragmatics was accompanied by formal and computational work from the start, with two main strands of approaches emerging at opposite ends of a spectrum. On one end of the spectrum, *grammaticalism* focuses on language as an abstract, computationally encapsulated system [MF: refs]. Grammaticalism strives to give formally precise explanations for how pragmatic inferences (focusing almost exclusively on interpretation) could arise in a language-specific computational module [MF: refs]. On the other end of the spectrum, *probabilistic pragmatics* seeks explanations of pragmatic phenomena from the point of view that linguistic communication as a special form of social interaction, also taking aspects of domain-general cognition of language users into account [MF: refs]. Figure 1 gives a coarse-grained overview of the development of, in particular, probabilistic pragmatics as the culmination point of research that combined aspects of individual cognition, such as bounded rationality [MF: refs], and functional explanations based on optimal information transfer in communication [MF: refs].

While experimental data can mitigate between alternative explanations proposed in the grammaticalist tradition [MF: refs], I argue here that the more socio-cognitive approach adopted by probabilistic pragmatics is, if done right, much better suited to incorporate *both* linguistic theory *and* aspects of general cognition (such as perception, adaptation and decision-making), as well as

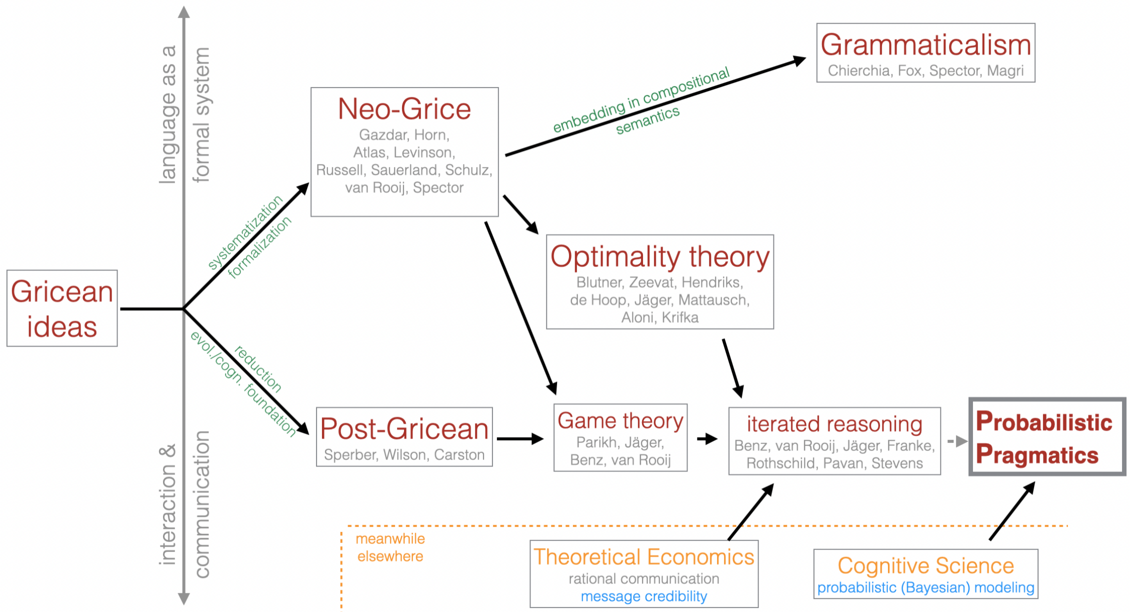

Figure 1: Genealogy of probabilistic pragmatics. [MF: describe what's pictured here (if you keep it; maybe make this figure and the text part of an InfoBox?)]

the concrete affordances of a conversational situation or an experimental task, into a unified computational model of the *data-generating process*, i.e., the (cognitive) processes by which individual data observations, such as concrete data points recorded from an experiment, come about. Seen in this light, computational modeling in pragmatics can serve a bridging function between language, linguistic cognition and domain-general aspects of information-processing and decision-making. The central argument of this position paper is, consequently, that a possible and highly productive future of experimental pragmatics in general consists in a dedicated *cognitive turn*, which seeks to explain language use as the intricate interplay between language-specific and domain-general cognition. Computational cognitive-pragmatic models plays a pivotal role in this cognitive turn, as they require detailed commitment about the role of linguistic theory (latent, established concepts) in the data-generating process, and the way that general cognitive factors shape linguistic behavior. As I will argue here, recent work in modelling-oriented experimental pragmatics has started to "go cognitive" in this sense, but that a more dedicated embrace of the cognitive turn is needed. Looking forward, I sketch possible future directions.

[MF: spell out paper overview] Section [MF: XYZ] introduces probabilistic pragmatics models. Section [MF: XYZ] elaborates on the idea of cognitive models acting as theory-driven statistical models for the data-generating process. It covers some main lessons learned from probabilistic pragmatic modeling of experimental data, and points to key open questions for the future. Section [MF: XYZ] then focuses on a high-level case study, namely probabilistic reasoning about common ground, as an illustrative example . . .

2 Probabilistic pragmatics

Notable early probabilistic approaches to pragmatics include pioneering work by Merin (1994, 1999), Zeevat (2014), and Russell (2012). A particularly prominent style of probabilistic modeling in pragmatics is the Rational Speech Act framework (Frank and Goodman, 2012; Scontras, Tessler, and Franke, 2021; Degen, 2023), the basic set-up of which was anticipated in decision-theoretic work by Anton Benz (Benz, 2006; Benz and van Rooij, 2007). What unites probabilistic approaches to pragmatics is, arguably, the idea that pragmatic interpretation is a form of inverse, or abductive, reasoning (Hobbs, Stickel, and Martin, 1993; Bunt and Black, 2000; Franke and Jäger, 2016): since it is natural to assume that a speaker's linguistic behavior (their choice of words, the fact that they spoke in the first place, their intonation, or speech-accompanying gesture, etc.) is at least partly determined by a latent, i.e., unobservable, mental state of the speaker, the main task of pragmatic interpretation is to reason backwards, as it were, about how relatively likely or unlikely different conceivable causes of the speaker's behavior are.

This type of reasoning is naturally modeled by Bayesian probabilistic inference. Given a probabilistic speaker policy $\pi(a \mid t)$, which specifies how likely a speaker is to perform an action $a \in A$ when in state $t \in T$, together with a prior distribution over how likely different states t are, the normatively correct way of updating these prior beliefs is given by Bayes' rule as follows:

$$P(t \mid a) = \frac{\pi(a \mid t)\, P(t)}{\sum_{t' \in T} \pi(a \mid t')\, P(t')} \tag{1}$$

To flesh out this basic—and, at least as a starting point, arguably natural—picture of interpretation with actual content, at least three things need to be specified: (i) the relevant set of states T; (ii) the relevant set of actions A; and (iii) the speaker's behavioral policy π. Consider, for instance, a standard case of referential communication. Here, the states T could be taken to be the potential referents in the shared perceptual context between speaker and listener, while the actions A could be a finite set of utterances or descriptions, identified, say, at the level of surface form, such as "the blue triangle." The speaker's policy π could then be defined on the basis of standard assumptions about how speakers are expected to behave in situations of the relevant kind, for example, as envisaged by Grice's Maxims of Conversation (Grice, 1975). In particular, for a simple case of referential communication, we might define a probabilistic policy in terms of a utility function that scores each action (utterance) a in a given state t based on a's truth-value in t (quality), the degree to which a conveys information about t as opposed to any other $t' \in T$ (informativity), and the cost $C(a)$ of performing a:

$$\mathrm{U}(a, t) = \mathrm{Informativity}(a) + \mathrm{Truth}(a, t) + \mathrm{Cost}(a) \tag{2}$$

The speaker's policy is then defined as (approximately) optimal behavior for this utility function, for instance as implemented via the softmax function (Franke and Degen, 2023):

$$\pi(a \mid t) = \frac{\exp \alpha\, U(a, t)}{\sum_{a' \in A} \exp \alpha\, U(a', t)} \tag{3}$$

The standard formulation of the RSA speaker policy, as given by Frank and Goodman (2012) and much subsequent work, conforms to this general utility-optimization picture, but is usually stated in terms of a speaker optimizing their linguistic expressions relative to a literal listener.

Since this literal listener is typically construed as a context-blind, unstrategic dummy, however, it is more useful to think of a probabilistic pragmatic model, more generally, as consisting of two components:

- a speaker policy that seeks to approximate optimal behavior under some (numeric) utility defining "efficient" or "successful" communication; and
- a Bayesian listener, who inverts this speaker policy in order to infer likely explanations for the observed linguistic behavior.

Many interesting variations, extensions and applications of this general approach to probabilistic pragmatic modeling have been developed over the years. These often extend the factors that feed into the speaker's utility function, e.g., to include aspects such as politeness or social meaning (e.g. Burnett, 2019; Yoon et al., 2020; Carcassi and Franke, 2023), or other aspects such as argumentativity (Carcassi, Wang, et al., 2026). [MF: insert more references] Another common extension is the inclusion of multiple latent variables for the listener to infer, each of which may shape the speaker's policy in different functional ways. Examples are the inclusion of a question under discussion (Kao et al., 2014), comparison classes [MF: refs], and more [MF: insert more examples]. All of these mentioned works and many others tightly link up with experimental data, either for testing categorical hypotheses supported by particular models, or, more frequently, to assess the quantitative fit of one or several models. If done right, this can be one of the assets that probabilistic modeling approaches bring to experimental pragmatics, as argued more extensively in the next Section 3.

Some models and applications also explore, or even require, the higher-order iteration of speaker policies and listener interpretations (e.g. Franke and Jäger, 2014; Bergen, Levy, and Goodman, 2016; Frank, Emilsson, et al., 2016). The general idea is that a speaker may optimize their linguistic behavior to be informative (or otherwise optimized) for a specific listener interpretation behavior that itself is optimized to the speaker's production policy. It then becomes an empirical question how much social reasoning is routinely employed by language users in general, or how much iterated perspective-taking individual language users are capable of [MF: refs]. However, from a resource-rational perspective on efficient cognitive processing (Lieder and Griffiths, 2019), we should expect to see meta-cognitive allocation of reasoning effort to be flexibly employed on-demand — a perspective that will be centrally spelled out in Section 4.

3 Theory-driven probabilistic modeling

- modeling in general:
    - all models are wrong; some are useful
    - even if all models are wrong, we should prefer models that are less wrong in the sense of Isaak
    - so how our (less) wrong models useful?
    - while for data: garbage in — garbage out
    - for models: more assumptions in — more conclusions out
        - e.g., causal modeling
        - reasonable assumptions in; interesting conclusions out
    - all models contain some assumptions about the **data-generating process**
    - more assumptions about the nature of that process → more information gained from the data

4 Towards the cognitive turn

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